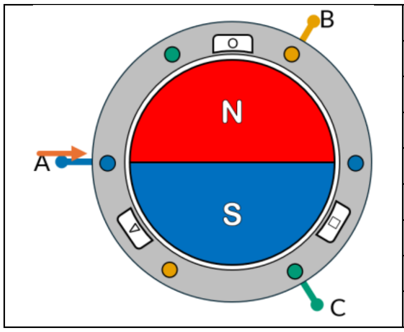
Recommended Assessment
Brushless DC Motor Lab

Hall Sensor Exploration

1. Based on the image provided in the lab, which hall sensor corresponded to the hall sensors shown with a circle, triangle and square? Fill out the table provided in the lab with the sequence in which the hall sensors get turned on as the motor rotates clockwise.



Hall Sensor 1	Hall Sensor 2	Hall Sensor 3

2. How many different states does the hall sensor goes through for a rotation of the magnet? How many times did that sequence repeat itself when rotating the motor manually for 1 full rotation? How many pole pairs are in the motor provided?

3. How many encoder counts are read in 1 mechanical rotation? Do encoder counts increase or decrease when rotating the motor clockwise? and counterclockwise?

4. How would you convert the encoder output from counts to radians?

Driving a BLDC Motor

5. Determine the control sequence needed to rotate the motor clockwise based on the hall sensor readings as shown in the lab. Use the table provided. For the decimal representation of the values, assume Hall Sensor 1 is the most significant bit. For each hall sensor state, note what phase should be High (voltage in), Low (voltage out) and Off (floating), using 1 for High, -1 for Low and 0 for Off.

Hall Sensor 1	Hall Sensor 2	Hall Sensor 3	Hall Sensors binary - decimal	Phase A	Phase B	Phase C

6. How would you convert your motor speed from counts/s to rads/s or to rpm?

7. How does changing the speed for a BLDC motor differs from changing the speed in a DC motor?

8. What is the minimum amplitude at which the motor starts turning. What is that in volts?

9. Plot the speed of the motor relative to the applied duty cycle. Does it match the speeds in the datasheet?